

PHILIP, SD, USA

WMO: 726516

Lat: 44.051N

Lon: 101.601W

Elev: 672

StdP: 93.50

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 24024

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-19.9	-26.4	0.4	-22.6	-23.5	0.5	-19.6	15.5	-2.3	14.2	-1.0	2.9	270	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	37.8	20.5	35.8	20.4	33.4	19.9	23.3	31.5	22.3	31.0	21.4	30.1	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	17.1	26.9	19.8	15.8	25.9	18.8	14.8	25.2	72.6	31.3	69.0	31.0	65.6	30.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.7	11.7	10.3	DB	-29.0	40.9	3.4	2.5	-31.4	42.7	-33.4	44.2	-35.3	45.6	-37.8	47.4
			WB	-29.0	24.9	3.0	1.1	-31.2	25.7	-32.9	26.3	-34.6	27.0	-36.8	27.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-4.8	-4.1	2.2	7.9	13.5	19.7	24.6	23.0	17.7	9.2	2.1	-4.5
	DBStd	11.85	7.26	7.75	6.99	5.61	4.72	4.19	3.68	3.71	5.04	5.35	6.20	7.00
	HDD10.0	1997	458	396	251	101	19	0	0	0	3	78	242	449
	HDD18.3	3907	716	629	499	313	160	32	2	6	71	286	486	707
	CDD10.0	1618	0	1	11	39	129	291	451	403	235	53	6	0
	CDD18.3	487	0	0	0	2	12	73	195	151	52	3	0	0
	CDH23.3	6883	0	0	15	73	251	1037	2578	1951	874	101	4	0
(13)	CDH26.7	3454	0	0	2	18	82	463	1424	1020	412	32	1	0
(14) Wind	WSAvg	4.4	4.4	4.4	4.8	5.2	4.9	4.3	4.0	4.0	4.2	4.5	4.2	4.3
(15) Precipitation	PrecAvg	390	9	11	22	42	72	76	44	40	28	33	12	9
	PrecMax	566	23	33	66	100	157	143	111	110	102	105	43	30
	PrecMin	174	0	0	0	9	7	8	1	8	0	0	0	0
	PrecStd	97	5	8	15	23	45	39	32	23	27	31	10	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	19.7	25.6	29.2	32.5	38.4	41.7	39.5	37.5	30.5	23.0	15.8
		MCWB	6.8	8.8	11.9	14.0	18.9	20.7	20.9	20.0	19.0	15.3	10.8	7.2
	2%	DB	10.9	14.1	21.3	25.6	28.9	34.3	38.3	37.1	34.1	26.2	18.8	11.2
		MCWB	4.9	6.2	10.3	13.5	17.7	20.6	21.0	20.3	18.2	13.7	9.3	4.9
	5%	DB	7.5	10.2	17.5	22.1	26.1	31.7	36.1	34.7	31.4	22.6	15.7	7.9
		MCWB	2.9	4.4	8.6	11.9	16.4	20.1	21.1	19.9	17.8	12.3	7.8	3.1
	10%	DB	5.2	6.9	13.8	18.6	23.3	29.0	33.7	32.4	28.5	19.2	12.2	5.1
		MCWB	1.8	2.7	6.9	10.2	15.0	19.0	20.5	19.6	16.8	11.1	6.1	1.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.1	12.9	15.5	21.1	24.3	25.0	23.5	21.5	16.6	11.8	7.5
		MCDB	14.8	18.4	22.3	25.9	28.7	30.5	32.8	32.6	31.1	27.3	21.8	14.6
	2%	WB	5.0	6.8	10.8	13.9	19.0	22.4	23.7	22.5	20.2	14.8	9.6	5.1
		MCDB	10.4	13.6	19.9	23.7	26.5	30.4	31.9	31.7	29.7	23.8	18.2	10.6
	5%	WB	3.3	4.6	9.1	12.5	17.3	21.2	22.7	21.6	19.0	13.1	7.9	3.3
		MCDB	7.1	10.0	17.2	21.1	24.7	29.6	31.1	30.5	28.4	20.7	15.1	7.8
	10%	WB	1.8	2.7	7.2	10.8	15.7	20.1	21.9	20.7	17.9	11.7	6.3	1.6
		MCDB	4.8	6.5	13.7	17.6	21.9	27.9	30.6	29.0	26.5	18.5	11.8	4.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.2	14.1	14.3	14.1	14.8	15.8	16.2	16.6	14.9	14.3	12.4
		MCDBR	15.9	18.6	21.2	21.2	18.8	19.0	19.0	19.8	21.3	21.2	20.0	16.5
		MCWBR	10.6	11.5	11.9	10.8	9.3	7.6	5.9	6.7	8.2	10.4	11.6	11.0
	5% WB	MCDBR	15.2	17.7	20.3	19.4	17.1	16.6	15.9	16.5	17.8	19.0	19.3	16.0
		MCWBR	10.4	11.1	11.7	10.2	9.0	7.9	6.5	6.6	7.5	9.7	11.4	10.9
(40) Clear-Sky Solar Irradiance	taub		0.260	0.266	0.288	0.329	0.351	0.368	0.386	0.379	0.342	0.296	0.268	0.257
	taud		2.555	2.539	2.529	2.402	2.412	2.393	2.377	2.379	2.460	2.564	2.554	2.542
	Ebn at Noon		894	946	960	935	919	901	879	874	885	895	874	862
	Edh at Noon		64	78	90	111	114	117	117	113	96	75	64	59
(44) All-Sky Solar Radiation	RadAvg		1.89	2.88	3.99	5.01	6.01	6.86	7.00	5.95	4.71	3.10	2.11	1.59
	RadStd		0.13	0.21	0.31	0.32	0.57	0.47	0.27	0.35	0.31	0.34	0.18	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-1.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page